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RESEARCH ARTICLE

A game-theoretic approach to designing carbon regulations in a duopoly freight transportation market: Road and multimodal road-rail competitive systems

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Abstract

Since the transportation industry produces remarkable carbon emissions worldwide, governments aim to curb these emissions by implementing different carbon regulations. However, little is known regarding how the heterogeneity of transportation systems should be exploited to design carbon regulations. To fill this research gap, the present article formulates carbon cap-and-trade and carbon offset regulations to reduce total carbon emissions produced by road and multimodal road-rail freight transportation systems (FTSs) in a duopoly market; the latter is regarded as a green, energy-efficient transportation mode. A novel procedure is suggested to allocate initial carbon caps that is a hybrid of both benchmark and grandfathering methods. The procedure allows the government to exploit the energy efficiency of the multimodal system, when targeting a reduction of total carbon. Then, a game-theoretic approach is adopted to implement the mentioned carbon regulations. A government, as a Stackelberg leader, maximizes a social welfare function containing economic, social, and environmental dimensions. Under the cap-and-trade regulation, a Nash bargaining process is proposed to trade carbon permits between the FTSs, as the followers, for updating their initial caps. The equilibrium outputs of the two mentioned carbon regulations and a carbon tax regulation are compared. The findings based on an experimental analysis suggest that the cap-and-trade (carbon offset) is the optimal and energy-efficient regulation from the social (economic or environmental) perspective. In terms of policy implications, our findings indicate that the development of a marketplace infrastructure for trading carbon permits is not justifiable under the economic and environmental perspectives of the government.

Keywords Cap-and-trade · Tradable carbon permit · Carbon offset · Carbon tax · Freight transportation · Game theory

Nomenclature

Parameters

i Index of freight transportation system (FTS_i, $i=u$, a). $i=u$ refers to the road system; $i=a$ refers to the multimodal road-rail system.

j Index of carbon regulation, $j=$ OF denotes carbon offset; and $j=$TR denotes cap-and-trade.

k Index of scenarios for the FTSs' reactions to the carbon regulations, $k\in\{SC_{V,V}, SC_{V,C}, SC_{C,V}, SC_{C,C}\}$. The first and second elements of the subscript refer to behaviors of FTS_u and FTS_a, respectively. V and C refer to violation and commitment to the updated cap, respectively.

$\theta_1(\theta_2)$ Fuel consumption rate of trucks (trains) for carrying 1 unit of demand within 1 unit of distance ($\frac{lit}{\text{ton}\cdot\text{km}}$), $\theta_1>\theta_2>0$

A_i The market base for the demand of FTS_i (ton), $A_i>0$

α_{pi} Self-price sensitivity of the demand for FTS_i, $\alpha_{pi}>0$

β_p Cross-price sensitivity of the demand, $0<\beta_p<\alpha_{pi}$

D_i Distance between an origin and a destination in FTS_i (km), $D_i>0$

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d Ratio of total railway in distance between the origin and the destination in multimodal road-rail transportation, $0 < d \leq 1$

e_{i0} Average value of CO_2 emissions produced by FTS_i for carrying 1 unit of demand from the origin to the destination $\left(\frac{\text{ton } CO_2}{\text{ton}}\right)$

Q_{i0} Demand of freight transportation for the FTS_i in the status quo (i.e., without any carbon regulation) (ton)

e_0 Average value of CO_2 emissions produced in the transportation industry, involving FTS_u and FTS_a $\left(\frac{\text{ton } CO_2}{\text{ton}}\right)$, $e_0 = \frac{e_{u0} \times Q_{u0} + e_{a0} \times Q_{a0}}{Q_{u0} + Q_{a0}}$

f Fuel price $(\frac{\$}{lit})$, $f > 0$

$\overline{c}_i$ Total cost imposed on FTSs for carrying 1 unit of demand from the origin to the destination excluding fuel cost $(\frac{\$}{\text{ton}})$, $\overline{c}_i > 0$. $\overline{c}_a$ includes transloading cost.

c_i Transportation cost imposed on FTS_i including fuel cost $\left(\frac{\$}{\text{ton}}\right)$, $c_i > 0$

$\varnothing$ Environmental concern parameter, e.g. the environmental damage of 1 unit of CO_2 emission $(\frac{\$}{\text{ton } CO_2})$

F Carbon trading price $\left(\frac{\$}{\text{ton } CO_2}\right)$

λ_1 Importance coefficient of economic considerations (i.e., FTSs' profitability) in the government's social welfare function, $0 \leq \lambda_1 \leq 1$

λ_2 Importance coefficient of social considerations (i.e., consumer surplus) in the government's social welfare function, $0 \leq \lambda_2 \leq 1$

λ_3 Importance coefficient of environmental considerations (i.e., carbon emissions) in the government's social welfare function, $0 \leq \lambda_3 \leq 1$

Decision variables

γ Carbon emission reduction coefficient determined by the government, $\gamma \in (0, 1)$

ψ Government penalty imposed to an FTS due to its violation of a carbon regulation $(\frac{\$}{\text{ton } CO_2})$

p_{ijk} Transportation price for carrying 1 unit of demand from an origin to a destination in FTS_i under regulation j and scenario k, $\left(\frac{\$}{\text{ton}}\right)$

s_{trade} Amount of carbon traded between the FTSs in the bargaining process under the cap-and-trade regulation (ton CO_2)

Dependent variables

q_{ijk} Transportation demand of FTS_i under regulation j and scenario k (ton)

$\overline{s}_i$ Initial carbon cap allocated to FTS_i by the government (ton CO_2), $\overline{s}_u = \gamma e_0 Q_{u,0}$, $\overline{s}_a = \gamma e_0 Q_{a,0}$

y_{ijk} The final amount of surplus carbon permits of FTS_i under regulation j and scenario k ($y_{i,j,k} < 0$ denotes shortage of carbon permits) (ton CO_2)

π_{ijk} Profit of FTS_i under regulation j and scenario k ($)

SW Social welfare function

Introduction

Motivation

Nowadays, the concept of *sustainability* is suggested by concerned activists and researchers to tackle some real-world challenges that involve interdependent *economic*, *environmental*, and *social* issues (Zarei and Rasti-Barzoki 2022). Global value chains are concerned with the environmental sustainability from energy consumption and the associated carbon emission (Awan et al. 2022a; Awan et al. 2022b). On the one hand, this emission is linked to the environmental deterioration of climate change and global warming (Akhmat et al. 2014; Knippertz et al. 2015). For example, climate scientists evaluate the social cost of carbon as equal to $51 per ton of CO_2; however, the Environmental Protection Agency (EPA) estimates that the mentioned cost is about $190 in 2022 (Asdourian and Wessel 2023). On the other hand, it may be inevitable to tolerate a certain amount of energy consumption and carbon emission to generate economic sustainability for firms and also social sustainability for consumers (Tamannaei et al. 2021b).

To avoid the mentioned climate change damages, it is vital to curb carbon emissions worldwide. In this regard, carbon regulations are designed to reduce carbon emissions, which enhance environmental sustainability (Audy et al. 2012). The main regulations suggested by policymakers to curb carbon emissions are carbon cap-and-trade, carbon tax, carbon offset, and carbon cap. Carbon cap-and-trade and carbon tax regulations are implemented widely in different countries (Liu et al. 2021; Xu et al. 2016b). In the cap-and-trade regulation, the government often allocates a carbon cap (hereinafter cap) to each firm, and firms can trade some tradable carbon permits (hereinafter permit) with each other in a carbon market (Gu et al. 2019; Malladi and Sowlati 2020a; Wagner 2013). If a firm produces lower (higher) emissions than its cap, it sells (purchases) the surplus (shortage of) permits in the market; this trade updates its cap. In the cap-and-trade and carbon offset regulations, if a firm violates its cap, it has to pay a penalty for extra emissions. In contrast to the cap-and-trade regulation, firms cannot sell the surplus permits in the carbon offset regulation (Malladi and Sowlati 2020b; Sherafati et al. 2020). Therefore, in contrast to the carbon offset, the cap-and-trade regulation requires a marketplace infrastructure, like an online platform, to realize the carbon trades.

Finally, the carbon tax regulation can be considered as a special case of the cap-and-trade regulation in which no permit is assigned to a firm, and its cap is also equal to zero. Hence, this zero cap is always violated under the tax regulation and a penalty (i.e., tax) is imposed on the firms for producing every unit of carbon emission (Chen et al. 2020).

The objective of this article is to design cap-and-trade, carbon offset, and carbon tax regulations for a duopoly freight transportation market that involves a road and a multimodal road-rail transportation system. The necessity of the research originates from the significant role of transportation in producing carbon emissions. In fact, according to the World Economic Forum, freight transportation causes to emit 5% of the total world carbon emissions (Chao et al. 2019). Based on EPA, transportation produced about 26% of total carbon emissions of the US in 2014. In addition, EPA announces that the aviation transportation produces 9% of total carbon emissions generated by the transportation industry in 2017 (Chao et al. 2019). Accordingly, the crucial mission of the governments is to reduce carbon emissions related to transportation by implementing different carbon regulations (Hermeling et al. 2015; Kang et al. 2022).

Studying carbon emission in freight transportation systems is also relevant from a sustainability perspective. In contrast to conventional systems, a sustainable freight transportation system not only reduces carbon emissions because of consuming less fossil fuels (Taheri and Tamannaei 2023; Tavassoli and Tamannaei 2020) but also enhances the economic prosperity (Consult 2020; Shaheen et al. 2022), as well as improving the equal accessibility of consumers to transportation systems (Black 2004).

Heterogeneity of road and rail freight transportation systems are significant from both the consumers' and the government's viewpoints. The road freight transportation system provides some economic and social advantages, including higher speed, flexibility, accessibility, and low maintenance cost. In contrast, the rail freight transportation system consumes less fuel and emits less carbon pollution than the road system. According to Van Essen et al. (2011), the average external costs of the rail system, including greenhouse gas emissions, noise, and accident are about 6.4 times lower than those for the road system. Therefore, the rail system is considered sustainable green transportation (Chen et al. 2017a; Tamannaei et al. 2021a; Tamannaei et al. 2021b; Tsai 2017).

According to the advantages of the road and rail systems, the multimodal road-rail system can be used for carrying cargoes to achieve sustainable goals. Moreover, road and multimodal road-rail systems are partially substitutable from the consumers' viewpoint and can compete with each other in a given market segment (Janic 2007; Tamannaei et al. 2021b). When designing carbon regulations, the government should pay attention to the green advantage of the multi-modal system, aiming to reduce the total carbon emissions of the transportation industry.

Theoretical gaps and research questions

While "Motivation" explains the practical relevance of the study, there are significant gaps in the existing literature with respect to the comprehensive aspects of carbon regulations in the transportation industry. On the one hand, although the existing literature has investigated the impacts of carbon regulations on the transportation industry (Micheli and Mantella 2018; Wangsa et al. 2020; Xu et al. 2016a), the heterogeneity of transportation modes with respect to their carbon emissions has not been addressed yet. On the other hand, the literature has concentrated on the government assessment of social welfare when studying carbon regulations (Anger and Köhler 2010; Khan et al. 2016; Tang et al. 2017). However, the impacts of economic, social, and environmental aspects on the effectiveness and optimality of carbon regulations in transportation markets have not been addressed. Studying these three impacts can help us achieve a comprehensive sustainability assessment for carbon regulations.

Some scholars have compared the effectiveness of different carbon regulations to reduce carbon emissions (Chen et al. 2020; Flachsland et al. 2011; Hu et al. 2020; Zhao et al. 2020). The present article contributes to the literature not only by comparing the effectiveness of carbon regulations, but also by assessing their optimality from the economic, social, and environmental aspects. Such comprehensive assessment is conducted for cap-and-trade, carbon offset, and carbon tax regulations in a duopoly freight transportation market containing road and multimodal road-rail systems. In this regard, the article aims to address the following research questions in the context of the duopoly freight transportation market:

Question 1. *Among the carbon tax, carbon offset, and cap-and-trade, which carbon regulation is effective for reducing carbon emissions?*

Question 2. *How does the government determine the optimal carbon regulation among the three regulations?*

Question 3. *How does the green advantage of the multimodal system affect the selection of the optimal regulation?*

Question 4. *How does the government's perspective affect the selection of the optimal carbon regulation?*

Question 5. *Under which conditions is the development of a marketplace infrastructure for trading carbon permits justifiable for the government?*

Outline

The rest of this paper proceeds as follows. "Theoretical background" provides the theoretical background of the article. "Literature review" reviews the related literature on

carbon regulations and transportation systems. "Problem description" explains the problem description. In "Methodology" (methodology), the initial carbon caps, demands, and social welfare function are formulated, followed by the development of carbon regulations. An experimental analysis with sensitivity analysis is presented in "An experimental analysis." "Discussions and implications" provides discussions and practical implications based on the experimental analysis. Finally, conclusions and suggestions for future research are elaborated in "Conclusions."

Theoretical background

Three aspects of sustainability, i.e., economic, social, and environmental, mutually affect each other. Accordingly, governments and firms encounter an inherent trade-off between these aspects, when making their decisions. There is a long-standing debate in the literature when addressing this trade-off.

Improving energy efficiency is the first approach to address the mentioned trade-off. Energy efficiency of a firm or industry is defined as its total output divided by its total energy consumption (Linares and Labandeira 2010; Patterson 1996). Hence, a firm is more energy efficient if it consumes less energy to produce the same amount of outputs in the form of products or services. The target of enhancing energy efficiency can be implemented using some green technologies that improve the production processes, utilize renewable energy sources, and exploit green raw materials (Zhang et al. 2019). Economic and social sustainability related to a firm depends strongly on the level of its output (Tamannaei et al. 2021b). Thus, given a fixed level of output, a firm with higher energy efficiency is more environmentally sustainable, without an adverse effect on economic and social sustainability.

Another approach to maintaining an appropriate trade-off between the three aspects of sustainability is the adoption of carbon regulations by governments. As mentioned earlier, enhancing energy efficiency *directly* targets the reduction of energy consumption and carbon emission, by using some green technologies. In contrast, carbon regulations aim to *indirectly* control the level of carbon emissions using some incentive and punitive rules implemented on the consequences of the firms' actions. Some scholars imply the necessity of carbon regulations. They indicate that enhancing energy efficiency may not be an effective solution to curbing total carbon emissions, unless it is combined with a carbon regulation (Herring 2006; Tamannaei et al. 2021b). This is because higher energy efficiency induces higher total outputs, which may increase total energy consumption and consequently higher carbon emission. The phenomenon is called rebound effects (Agrawal and Bellos 2017). In this regard, carbon regulations can mitigate the rebound effect phenomenon since they can control the level of the firms' outputs (Tamannaei et al. 2021b).

Another research stream indicates the relationship between the two approaches. Ngo (2022) and Wang et al. (2022) believed that the cap-and-trade, carbon tax, and carbon offset regulations, as market-based mechanisms, can increase the energy efficiency indicator. On the other hand, carbon regulations contribute to carbon emission reduction and significantly impact energy efficiency (Nam and Jin 2021; Xu and Xu 2022). The rationale behind this relationship is that carbon regulations can improve the firms' utility of employing green technology (Kushwaha et al. 2020; Yang et al. 2020a; Zhang et al. 2019), thereby enhancing energy efficiency indirectly. Hence, our first hypothesis is as follows:

H1. Implementing a carbon regulation is positively related to improving energy efficiency.

There are some real-world examples regarding the implementation of carbon regulations in the transportation industry. Since 2012, the European Commission has included aviation transportation to the EU emissions trading system, i.e., a type of carbon cap-and-trade regulation. All airlines operating in Europe are required to reduce their emissions given their caps, regarding their capability to trade their allowances (European_Commission 2022). The European Commission also imposes a carbon tax on the freight shipping industry. Today, the value of carbon tax is equal to \$95 per ton of CO_2. Moreover, analyses predict that the tax of \$200 per ton of CO_2 will be required to decarbonize the freight shipping industry by 2050 (Gerretsen 2022).

We suggest a new definition for the effectiveness of a carbon regulation. A carbon regulation is defined to be effective if it provides less total carbon emissions than the status quo without any regulation. An existing research stream aims to compare the effectiveness of different carbon regulations. Chen et al. (2020), Hu et al. (2020), and Flachsland et al. (2011) compare the carbon emission reduction of cap-and-trade and carbon tax regulations. They concluded that although both regulations are effective, the cap-and-trade regulation reduces emissions more than the carbon tax. Accordingly, the cap-and-trade regulation remarkably improves the environmental aspect of sustainability. Hence, we hypothesize that:

H2. The cap-and-trade is the preferred regulation to implement.

Literature review

The present article is closely related to two research streams. The first stream reviews the literature addressing the effects of carbon regulations on the transportation industry. The rationale behind such a review is that the present article studies two freight transportation systems under four carbon regulations.

The second research stream reviews the literature addressing the effects of carbon regulations on motivating enterprises to adopt green technology. The present article is relevant to the second stream in a sense that it addresses competition between road and multimodal road-rail transportation systems; the latter can be regarded as a green technology (Chen et al. 2017a; Tamannaei et al. 2021a; Tamannaei et al. 2021b). Since the government aims to benefit from the rail transportation system to achieve environmental sustainability, the general literature on governance for green technology adoption should be critically reviewed.

The effects of carbon regulations on the transportation industry

Researchers aimed to curb carbon emissions in the transportation industry by implementing different carbon regulations. Xu et al. (2016a) considered a bi-level multi-objective problem to reduce carbon emissions in the aviation transportation industry under the cap-and-trade regulation. They analyzed conflicts between the government and aviation transportation companies, about carbon emission reduction and air transportation demands. Micheli and Mantella (2018) developed an inventory routing problem for heterogeneous fleets, by considering demands uncertainty and different carbon regulations, i.e., cap-and-trade, carbon tax, and carbon offset. They compared operational configurations of the carbon regulations from economic and environmental implications. They concluded that for the same carbon reduction, the carbon tax increases the transportation cost more than the cap-and-trade. Wangsa et al. (2020) and Jiang et al. (2015) investigated an inventory optimization problem considering transportation costs under carbon tax and cap-and-trade regulations from the economic and environmental implications. They concluded that the loading and unloading activities have a significant role in minimizing total cost of the inventory system under both regulations. According to Wang et al. (2015), closed and open cap-and-trade could reduce total carbon emissions produced by container and dry bulk shipping sectors; the two sectors are considered as heterogeneous and competitive. Sheu and Li (2014) investigated the relationship between consumer behaviors and investment in green aviation in a duopoly market. They concluded that airlines should stimulate adopting green aviation as soon as possible.

According to the literature review on the first research stream, three research gaps are identified in the literature, which have not been addressed yet, to the best of our knowledge:

Research gap 1: Researchers have studied the competition of heterogeneous transportation systems under carbon regulations (Sheu and Li 2014; Wang et al. 2015; Xu et al. 2016a). Such heterogeneity is considered only as a driver for the competition, and its relationship with carbon regulations is remained unknown in the existing literature. Accordingly, there is no implication regarding how this heterogeneity should be employed to design carbon regulations with the goal of environmental considerations.

Research gap 2: The existing literature has considered the effects of carbon regulations on transportation costs incurred by consumers of transportation services (Jiang et al. 2015; Micheli and Mantella 2018; Wangsa et al. 2020). The social welfare of a society is more comprehensive than the transportation costs. Even so, effects of carbon regulations on the comprehensive social welfare have not been investigated.

Research gap 3: There are few studies in which different carbon regulations have been compared (Micheli and Mantella 2018; Wang et al. 2015). For instance, Micheli and Mantella (2018) concluded that the carbon cap-and-trade is the preferred regulation when compared with the carbon tax regulation. However, the effects of the government's perspectives on the optimality of a carbon regulation has remained unknown in the existing literature.

The effects of carbon regulations on green technology adoption

Some studies have concentrated on the adoption of green technology in industries other than transportation, e.g., manufacturing and electricity generation. Wang et al. (2018) investigated the direct and indirect effects of the cap-and-trade regulation on suppliers and manufacturers. They concluded that an enterprise with lower carbon intensity can easily resist on the pressures of direct and indirect effects and is able to increase its outputs. Zhang et al. (2019) designed an evolutionary game with government intervention and the manufacturers' choice between adopting green technology and purchasing extra carbon credits. The researchers concluded that the manufacturers' equilibrium decisions to adopt green technology are positively correlated with the penalty of violation. Liu et al. (2021) developed a bi-objective, nonlinear model that oligopolistic manufacturers decide to select the best green technology with respect to maximizing their profits. The researchers concluded that if the environmental concern is low, the cap-and-trade is ineffective to promote investment in green technology. Cao et al. (2017) designed a Stackelberg game in which a government, as the leader, can adopt a cap-and-trade policy or a Low Carbon Subsidy Policy (LCSP) to decrease carbon emissions. In response to the selected policy, manufactures, as the followers, determine their product prices and the reduction level of their carbon emissions. If the environmental damage coefficient is higher than a certain threshold, the cap-and-trade policy was found to be superior to the LCSP. Yang et al. (2020b) investigated investment in green technology using the benchmark and grandfathering as the two common methods for allocating

initial caps in the cap-and-trade and carbon offset regulations. According to the results, the benchmark is more effective than grandfathering to motivate enterprises for investing in green technology. Some studies conducted on the electric power sector indicated that under the cap-and-trade regulation, investment in renewable and green energies is capable to reduce carbon emissions (Chen et al. 2021; Delarue and Van den Bergh 2016; Yu et al. 2016; Yu et al. 2021).

According to the literature review on the second research stream, one research gap is identified in the literature, which has not been addressed yet, to the best of our knowledge:

Research gap 4: Few studies have analyzed the effect of the penalty on motivating manufacturers to invest in green technology. For example, Zhang et al. (2019) concluded that investment in green technology has a positive correlation with the penalty of violation. To expand the knowledge on the penalty of violation, its effects on the agents' profits, either they invest in green technology or not, should be assessed.

Table 1 summarizes a synthesis of the related literature and its comparison with the present article. In this paragraph, the logic behind a column in Table 1 is explained comprehensively. First, the articles are categorized into research streams I ("The effects of carbon regulations on the transportation industry") and II ("The effects of carbon regulations on green technology adoption"). According to the literature review, the effects of carbon regulations on the "transportation industry" and "green technology adoption" are the two research streams. Scholars usually apply cap-and-trade, carbon offset, and carbon tax regulations in their studies to reduce carbon emissions. Under the cap-and-trade and carbon offset regulations, grandfathering and benchmark methods, as the two conventional methods, are used for allocating the initial caps to enterprises. Moreover, the present article proposes the bargaining process to trade permits between the FTSs. Since the effects of carbon regulations should be investigated based on the sustainability aspects, the three economic, social, and environmental considerations composing the sustainability concept are assessed. Finally, the last two columns indicate the methodologies adopted by studies. In this regard, the studies usually consider Nash and Stackelberg games as the common methodologies to implement carbon regulations.

Figure 1 indicates the distribution of articles that have investigated different carbon regulations and government perspectives mentioned in Table 1.

Main contributions

The present article contributes to the existing literature by investigating competition between heterogeneous transportation systems, with respect to their energy efficiency, under carbon regulations, i.e., cap-and-trade, carbon offset, and carbon tax. To fill research gap 1, the following contributions are provided in the article:

- To govern the pricing competition between road and multimodal road-rail freight transportation systems, carbon cap-and-trade and carbon offset regulations are customized.
- A novel procedure for allocating and trading carbon permits is proposed. The procedure allows the government to employ the green advantage of the rail transportation system to reduce total carbon emissions in the transportation industry.
- A significant characteristic of the mentioned procedure is a bargaining process for determining traded permits between FTSs. To formulate the bargaining process, a Nash bargaining method is adopted, which has not been applied in the existing literature.

Also, the following contributions are provided to fill the other research gaps:

- To fill research gaps 2 and 3, a game theoretic model is proposed in which the government determines its decisions using a social welfare function composed of three considerations: economic (FTSs' profits), social (consumer surplus), and environmental (the environmental damage of carbon emission).
- To fill research gap 4, the effects of violation penalty on motivating to invest in green technology and changing FTSs' profits are addressed.

To summarize, the present article compares three carbon regulations (i.e., cap-and-trade, carbon offset, and carbon tax) from different government perspectives. The results reveal that the binding cap-and-trade regulation is superior to the other ones if the social perspective is considered by the government. However, the carbon offset regulation is the best one from the economic or environmental perspective.

Problem description

The present article aims to address how different carbon regulations affect carbon emissions produced by the two rival freight transportation systems (FTSs), i.e., road and multimodal road-rail systems represented by FTS_u and FTS_a, respectively. More specifically, the article compares the carbon cap-and-trade regulation with the carbon tax and carbon offset regulations. FTS_u uses trucks to carry cargoes from an origin to a destination on a road network. In contrast, both trains and trucks carry the cargoes handled by FTS_a from that origin to that destination. Since the two mentioned FTSs

Table 1 The synthesis of the literature on carbon regulations

Article	Research stream		Carbon regulation			Carbon permits allocation method			Policy evaluation perspective			Methodology	
	(I) Transportation	(II) Green technology adoption	Carbon tax	Carbon offset	Carbon cap-and-trade	Grandfathering	Benchmark	Bargaining	Economic	Environmental	Social	Stackelberg game	Nash game
Xu et al. (2016a)	✓				✓	✓			✓			✓	✓
Micheli and Mantella (2018)	✓		✓	✓	✓	✓			✓	✓			
Wangsa et al. (2020)	✓		✓		✓				✓	✓			
Jiang et al. (2015)	✓				✓	✓			✓	✓			
Wang et al. (2015)	✓				✓	✓			✓	✓			
Chen et al. (2020)		✓	✓		✓	✓			✓	✓			
Zhang et al. (2019)		✓			✓	✓			✓	✓			
Yang et al. (2020b)		✓			✓	✓	✓		✓	✓			
Delarue and Van den Bergh (2016)		✓			✓	✓			✓	✓			
Sheu and Li (2014)	✓	✓			✓	✓			✓	✓	✓		
Cao et al. (2017)		✓			✓	✓			✓	✓	✓	✓	
Li et al. (2018a)		✓			✓	✓			✓	✓	✓	✓	
Du et al. (2016a)		✓			✓	✓			✓	✓			
Yu et al. (2021)		✓			✓	✓			✓	✓			
Chen et al. (2021)		✓			✓	✓	✓		✓	✓	✓		
Zhang et al. (2018)		✓			✓	✓			✓	✓		✓	✓
Du et al. (2016b)		✓			✓	✓			✓	✓			
Liu et al. (2021)		✓			✓	✓			✓	✓			✓
Anand and Giraud-Carrier (2020)		✓	✓		✓	✓			✓	✓	✓		✓
Wang et al. (2018)		✓			✓	✓			✓	✓		✓	✓
Yu et al. (2016)		✓			✓	✓			✓	✓			
Scott et al. (2020)		✓			✓	✓			✓	✓			
Chen and Nie (2020)		✓			✓	✓			✓	✓	✓	✓	
Lu and Sun (2021)		✓			✓	✓			✓	✓	✓		
Li et al. (2018b)		✓			✓				✓	✓		✓	
Suo et al. (2017)		✓			✓	✓	✓		✓	✓			
Our article	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

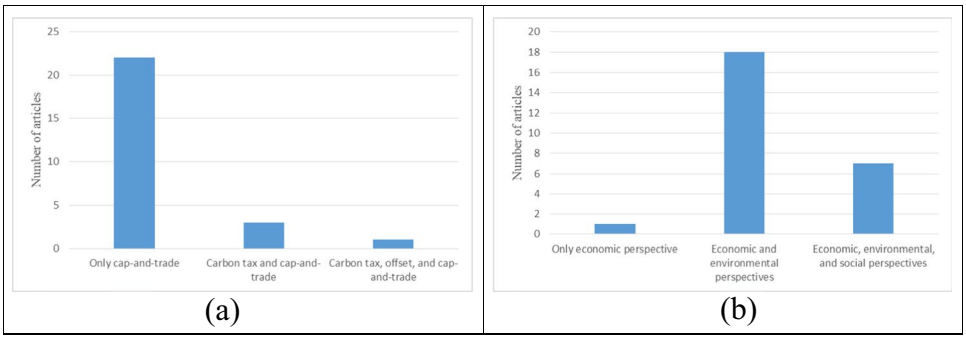


Fig. 1 The distribution of the articles that addressed: (**a**) carbon regulations and (**b**) government perspectives

have differences in some features, including affordability, safety, and environmental indicators, they can be considered rival freight transportation systems (Rategh et al. 2022; Tamannaei et al. 2021a; Tamannaei et al. 2021b). More currently, the services offered by FTS_a are superior to those of FTS_u considering energy consumption and carbon emission (Janic 2007). Hence, the government aims to exploit such superiority when designing its carbon regulations.

The agents and their decisions: A government allocates the initial caps to the FTSs ($\bar{s}_i$) and determines the penalty (ψ) for extra emissions to maximize social welfare. A bargaining process is proposed to trade permits between the FTSs in the cap-and-trade regulation. Each FTS sets its equilibrium price in a duopoly market considering a Bertrand-Nash competition framework.

Types of cap-and-trade regulation: Wang et al. (2015) suggested two approaches to implementing the cap-and-trade regulation in trading permits. If the regulation is an open system, each firm in an industry can trade with another firm in other industries (e.g., manufacturing, agriculture, and electricity generation). On the contrary, if the regulation is a closed system, firms in a specific industry trade permits only with each other. The present article considers a closed cap-and-trade implemented in the freight transportation industry for reducing total carbon emissions.

Allocating initial caps to the FTSs: The present article suggests a novel procedure to allocate initial caps that is a hybrid of grandfathering and benchmark methods. The benchmark method helps us determine the average carbon emissions in the transportation industry (e_0) based on the average carbon emissions (e_{u0} and e_{a0}) and demands (Q_{u0} and Q_{a0}) of the FTSs in the status quo (Zheng et al. 2020). Then, the government allocates the initial caps ($\bar{s}_u$ and $\bar{s}_a$) to FTS_u and FTS_a, respectively, by the grandfathering method. The allocating procedure can be beneficial for FTS_a as a sustainable green transportation system; because this sustainability can provide an opportunity for FTS_a to benefit from surplus permits. Accordingly, under the closed cap-and-trade regulation, FTS_a can sell its surplus permits to FTS_u, through the bargaining process, hoping to earn some profits. In this regard, the FTSs determine their equilibrium prices based on their updated caps. In contrast to the cap-and-trade regulation, the FTSs do not trade permits in the carbon offset regulation. Thus, their updated caps are equal to the caps allocated by the government. For the two mentioned carbon regulations, the FTSs can violate their updated caps by paying the penalty of violation.

Government's mission: the government determines the carbon emission reduction coefficient ($\gamma \in (0,1)$) and penalty of violation (ψ), considering the social welfare function (Ji et al. 2019; Xu et al. 2016b). The function contains the FTSs' profits, consumer surplus, and the environmental costs of carbon emissions. For each FTS, the cap allocated by the government is equal to γ multiplied by the total carbon emissions in the status quo normalized based on the average carbon emissions (e_0). Regarding the fact that an FTS should be committed to its updated cap, if the FTS produces more carbon emissions than the cap, it has to pay a penalty proportional to its extra emissions. This penalty is equal to ψ multiplied by the amount of extra emissions. For cap-and-trade and carbon offset regulations, the government determines the values of γ and ψ to maximize the value of the social welfare function.

FTSs' reactions to the carbon regulations: Considering the options of commitment and violation, different scenarios for the reactions of the FTSs can be regarded. An FTS can be committed to (C) or can violate (V) its updated cap. Accordingly, four scenarios can be defined as follows:

SC_{VV}: Both FTS_u and FTS_a violate their updated caps. It is equivalent to conditions $y_{ujSC_{V,V}} \leq 0$ and $y_{ajSC_{V,V}} \leq 0$.

SC_{VC}: FTS_u violates its updated cap, but FTS_a is committed to its updated cap. It is equivalent to conditions $y_{ujSC_{V,C}} \leq 0$ and $y_{ajSC_{V,C}} \geq 0$.

SC_{CV}: FTS_u is committed to its updated cap, but FTS_a violates its updated cap. It is equivalent to conditions $y_{ujSC_{C,V}} \geq 0$ and $y_{ajSC_{C,V}} \leq 0$.

SC_{CC}: Both FTS_u and FTS_a are committed to their updated caps. It is equivalent to conditions $y_{ujSC_{C,C}} \geq 0$ and $y_{ajSC_{C,C}} \geq 0$.

Determining equilibrium scenarios: in the carbon offset regulation, the FTSs do not trade permits with each other; so they react to this regulation based on their initial caps. We

further show in relation (8) that the initial cap for FTS_a is higher than its cap in the status quo, so FTS_a is committed to its cap in the carbon offset regulation. Hence, SC_{VV} and SC_{CV} are invalid scenarios in this regulation. Also, we further show that the government decreases the initial cap of FTS_u; then FTS_u encounters the shortages of permits. Therefore, for the carbon offset regulation, SC_{CC} is also invalid, and SC_{VC} is considered as the realized equilibrium scenario.

Regarding the FTSs' commitment or violation, the government can design two types of cap-and-trade regulations. The first regulation is designed so that both FTSs will have sufficient economic incentives to trade permits and to be committed to their updated caps; that is, SC_{CC} is the realized equilibrium scenario. This equilibrium scenario is entitled as the binding cap-and-trade regulation in which the FTSs do not have incentives to violate their updated caps. The rationale behind the government policy-making to realize SC_{CC} may be the scenario guarantees its long-term legal legitimacy. The second regulation is designed so that the realized equilibrium scenario is SC_{VV}. More specifically, the regulation provides incentives for the FTSs to trade permits, violate their updated caps, and to be charged by the government due to the violation. This equilibrium scenario is entitled as the non-binding cap-and-trade in which the FTSs may have incentives to violate their updated caps. The logic behind the non-binding cap-and-trade is that the FTSs may have more freedom of action than the binding cap-and-trade.

The decision timeline: Figure 2 shows the timeline of the problem and the decision variables related to each agent. Also, Fig. 3 indicates the agents' hierarchical decision timeline. At stage 1, the government, as the Stackelberg leader, determines the values of γ and ψ. At stage 2, the FTSs trade permits with each other through the bargaining process under the cap-and-trade regulation and jointly determine the value of s_{trade}. Evidently, stage 2 is eliminated under the carbon offset regulation. At stage 3, FTS_a and FTS_u, as the competing freight transportation systems (Tamannaei et al. 2021a; Tamannaei et al. 2021b), determine their equilibrium transportation prices (represented by p_{ijk}) using the Bertrand-Nash competition (Zhou and Lee 2009). The competition is

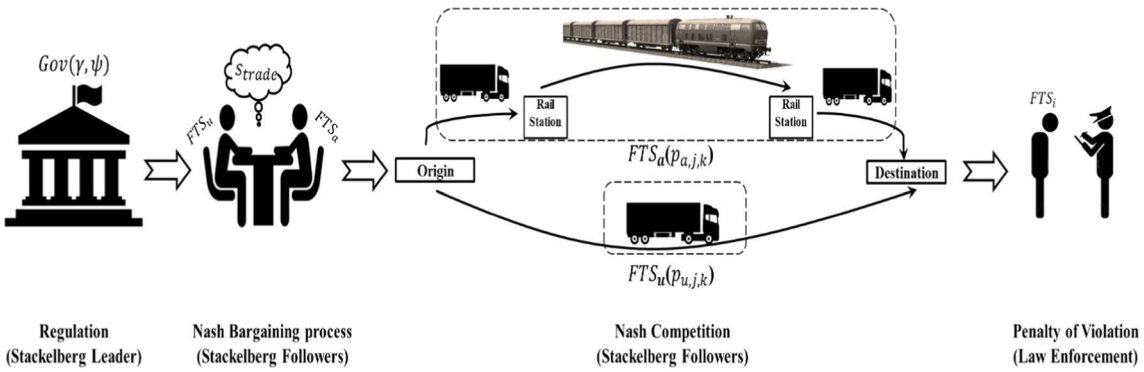


Fig. 2 Schematic view of the problem timeline

	Agents	Purpose	Agents' decisions
Stage 1	Government	Maximizing social welfare	**Stackelberg leader:** The government sets the values of carbon emission reduction coefficient (γ) and penalty (ψ) to maximize social welfare.
Stage 2	FTSs	Bargaining for trading permits	**Stackelberg followers:** FTS_u and FTS_a determine the traded carbon permits (s_{tarde}) through the bargaining process under the closed cap-and-trade.
Stage 3	FTSs	Pricing competition for profit seeking	**Stackelberg followers:** FTS_u and FTS_a determine their equilibrium transportation prices based on the Bertrand-Nash competition in the duopoly transportation market, i.e. $p_{u,j,k}$ and $p_{a,j,k}$, respectively.
Stage 4	FTSs	Realization of cap violation	**Stackelberg followers:** If an FTS produces more emissions than its cap, it must pay a penalty for its extra emissions.

Fig. 3 Explanation of the agents' decisions in a stage

significantly affected by the chosen carbon regulation (binding cap-and-trade, non-binding cap-and-trade, or carbon offset). The equilibrium transportation prices of the FTSs play a significant role in a customer's decision to select its desirable system; so, the demand of each FTS (q_{ijk}) is prominently affected by the prices (Chen et al. 2017b). Finally, at stage 4, if an FTS violates its updated cap, it pays penalties for the extra emissions. The backward induction methodology is adopted to solve the problem.

Assumptions: The following assumptions are considered for the problem.

Assumption 1. A linear demand function is considered for each FTS and is affected by the FTSs' prices regarding the self-price and cross-price sensitivities. The value of self-price sensitivity of demand is assumed to be greater than the value of cross-price sensitivity of demand, i.e., $\alpha_{pi} > \beta_p > 0$ ($i \in u, a$), indicating that a change in each FTS's price has more significant effect on its demand than a similar change in the rival price (Dan et al. 2012; Karray and Sigué 2015; Kurata et al. 2007; Wei et al. 2013).

Assumption 2. Due to the environmental superiority of FTS_a, the government assigns the initial cap of FTS_a (FTS_u) higher (lower) than its carbon emission under the status quo: $\bar{s}_a > s_{0,a} = e_{a0}Q_{a0}, \bar{s}_u < s_{0u} = e_{u0}Q_{u0}$.

Assumption 3. The fuel consumption rate of a truck is more than that of a train (i.e., $\theta_1 > \theta_2$) (Tolliver et al. 2013). According to Fig. 4, it is expected that total carbon emissions produced by FTS_a is less than that of FTS_u because the former consumes less fuel than the latter.

Methodology

The methodology section is devoted to formulate and solve the three carbon regulations: carbon offset, binding cap-and-trade, and non-binding cap-and-trade. In "Formulation of the costs, carbon caps, demands, and social welfare function," the costs, initial carbon caps allocated to the FTSs, demands of FTSs, and social welfare function are formulated. In Carbon offset regulation," SC_{VC} is developed as the realized equilibrium scenario for the carbon offset regulation. "Binding carbon cap-and-trade regulation" and "Non-binding carbon cap-and-trade regulation" present the binding and non-binding cap-and-trade regulations in which their realized equilibrium scenarios are SC_{CC} and SC_{VV}, respectively. The present article designs a novel procedure for the cap-and-trade regulation and compares the equilibrium outputs of the regulation with those of the carbon tax and carbon offset regulations.

In the cap-and-trade and carbon offset regulations, grandfathering and benchmark have been two conventional methods for allocating initial caps based on historical data of emissions (Yang et al. 2020b; Zheng et al. 2020). In the grandfathering method, a cap for total carbon production is allocated to a firm based on its individual historical emissions (He et al. 2017). However, the benchmark method allocates a cap for each unit of product based on the average value of emissions in the industry for that product (He et al. 2017; Yang et al. 2020b). Accordingly, the grandfathering (benchmark) method concentrates on reduction in the *total carbon emissions* (*emission per unit of product*).

Formulation of the costs, carbon caps, demands, and social welfare function

Costs imposed on an FTS are composed of fuel costs and other transportation costs such as transloading (Janic 2007). Equations (1) and (2) indicate the total transportation costs imposed on the FTSs:

$$c_u = \bar{c}_u + D_u\theta_1 f \tag{1}$$

$$c_a = \bar{c}_a + dD_a\theta_2 f + (1-d)D_a\theta_1 f \tag{2}$$

Average carbon emissions produced by FTS_a are calculated as below:

$$e_{a0} = e_{1a} + e_{2a} \tag{3}$$

We formulate initial caps allocated by the government for a given value of carbon emission reduction coefficient (γ) in

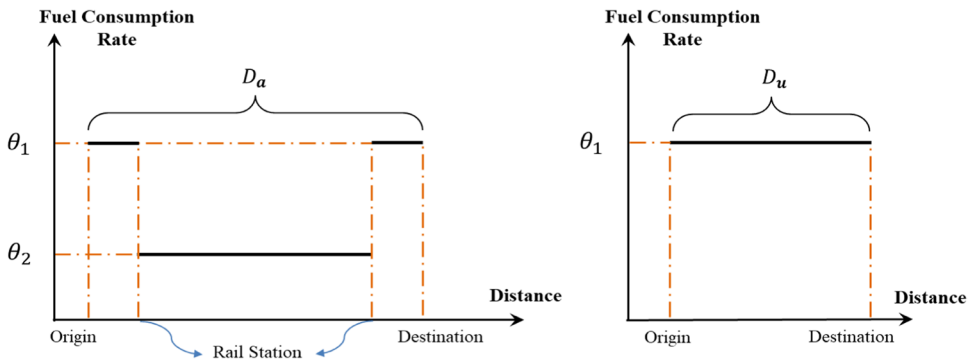


Fig. 4 Fuel consumption rates for FTS_a and FTS_u

a cap-and-trade or carbon offset regulation. In the status quo, the average CO_2 emissions (e_{i0}) and the demand values (Q_{i0}) are predetermined for the two FTSs ($i \in u, a$). Then, based on the benchmark method (Zheng et al. 2020), the average CO_2 emission for FTS_u, and FTS_a is calculated as follows:

$$e_0 = \frac{e_{u0} \times Q_{u0} + e_{a0} \times Q_{a0}}{Q_{u0} + Q_{a0}} \tag{4}$$

Since e_0 is an affine combination of e_{u0} and e_{a0} and relation $e_{a0} < e_{u0}$ holds, it is concluded that:

$$e_{a0} < e_0 < e_{u0} \tag{5}$$

Afterward, the grandfathering method (Zheng et al. 2020) suggests the following initial caps should be allocated to the FTSs, based on a given value of decision variable γ:

$$\bar{s}_u = \gamma \; e_0 \; Q_{u0} \tag{6}$$

$$\bar{s}_a = \gamma \; e_0 \; Q_{a0} \tag{7}$$

According to relations (5)–(7), it is concluded that:

$$\bar{s}_u < s_{0u} = e_{u0} Q_{u0}, \bar{s}_a > s_{0a} = e_{a0} Q_{a0} \tag{8}$$

According to relation (8), the grandfathering method in the cap-and-trade regulation is beneficial for FTS_a regarding that it can trade its surplus permits with FTS_u. This benefit for FTS_a originates from the green and sustainable nature of the rail system. Thus, FTS_a and FTS_u seem to have sufficient incentive to trade permits in the bargaining process.

According to the literature, the transportation demand (q_{ijk}) is assumed as a linear function of the FTS's price and the rival's price (Anderson and Bao 2010; Chen and Wang 2015; Choi 1991; Edirisinghe et al. 2011; Hafezalkotob 2018; Jeuland and Shugan 1983; Li et al. 2016; Pan et al. 2010; Xiao and Yang 2008; Xie 2015; Zhao et al. 2017). The effects of the prices are addressed by self-price and cross-price sensitivities, i.e., α_{pi} and β_p, respectively. Thus, the transportation demands of FTS_u and FTS_a are as follows:

$$q_{ujk} = A_u - \alpha_{pu} p_{ujk} + \beta_p p_{ajk} \tag{9}$$

$$q_{ajk} = A_a - \alpha_{pa} p_{ajk} + \beta_p p_{ujk} \tag{10}$$

The government aims to maximize the following social welfare function:

$$SW = \lambda_1 (\pi_{ujk} + \pi_{ajk}) + \lambda_2 \left(\int_{p_{ujk}}^{\frac{A_u + \beta_p \, p_{ajk}}{\alpha_{pu}}} (A_u - \alpha_{pu} \, x + \beta_p \, p_{ajk}) dx + \int_{p_{ajk}}^{\frac{A_a + \beta_p \, p_{ujk}}{\alpha_{pa}}} (A_a - \alpha_{pa} \, y + \beta_p \, p_{ujk}) dy \right) - \lambda_3 \varnothing (e_{u0} \times q_{ujk} + e_{a0} \times q_{ajk})^2 \tag{11}$$

As in He et al. (2017), Ji et al. (2019), Krass et al. (2013), and Xu et al. (2016b), the social welfare function is composed of three types of considerations: economic, social, and environmental. The FTSs' profits, i.e., $\pi_{ujk} + \pi_{ajk}$, can represent the economic consideration. The social consideration is represented by the total consumer surplus indicator. The indicator for a duopoly market is equal to $\int_{p_{ujk}}^{\frac{A_u + \beta_p \, p_{ajk}}{\alpha_{pu}}} (A_u - \alpha_{pu} \, x + \beta_p \, p_{ajk}) dx + \int_{p_{ajk}}^{\frac{A_a + \beta_p \, p_{ujk}}{\alpha_{pa}}} (A_a - \alpha_{p,a} \, y + \beta_p \, p_{ujk}) dy$, according to Besanko and Cui (2016), Hafezalkotob (2017), and Tamannaei et al. (2021b). Moreover, the quadratic form of total carbon emissions multiplied by the environmental concern parameter ($\varnothing$) can represent the environmental consideration according to He et al. (2017), Ji et al. (2019), and Xu et al. (2016b). Finally, the multipliers λ_1, λ_2, and λ_3 represent the importance coefficients of the three types of government perspectives. These multipliers depend strongly on the government's perspective and its priorities. One may wonder how these three coefficients can be estimated from a real-life situation. In this regard, discussion presented in the work of Kurth et al. (2017) can be helpful. Kurth et al. (2017) argue that the government's decision-making processes are diverse, complex, multidisciplinary, and sometimes involve contrasting objectives. Therefore, "a successful decision-making process (i.e., expeditious and credible) by government agencies requires stakeholder involvement," said Kurth et al. (2017). For example, the National Environmental Protection Act (NEPA) rule enforces government agencies to allow citizens to evaluate and weigh the environmental impact of federal projects. In this regard, λ_1, λ_2, and λ_3 can be set equal to the percentage of stakeholders involved in the government's decision-making process who have economic, social, and environmental concerns, respectively.

The government determines the equilibrium values of γ and ψ to maximize the social welfare function in each one of the three carbon regulations. The government evidently prefers to implement a carbon regulation that provides the higher value of the social welfare function and carbon emission reduction compared to the other regulations.

Carbon offset regulation

According to the backward induction methodology, the equilibrium price, demand, and profit of each FTS are obtained based on the Nash competition mentioned in stage 3. Then, the government's optimization problem to maximize the social welfare function is formulated, subject to the equilibrium outputs of stage 3.

In the carbon offset regulation, the FTSs do not trade permits with each other. Hence, they cannot update their initial

caps. Accordingly, their surplus permits under scenario k are equal to:

$$y_{uOFk} = \bar{s}_u(\gamma) - e_{u0} \times q_{uOFk} \tag{12}$$

$$y_{aOFk} = \bar{s}_a(\gamma) - e_{a0} \times q_{aOFk} \tag{13}$$

According to "Problem description," the government designs its carbon offset regulation so that scenario SC_{VC} is the realized equilibrium. Based on the definition of SC_{VC}, FTS_u encounters the shortage of permits and has to pay a penalty of $\left|\psi y_{uOFSC_{VC}}\right|$ for its extra emissions. In contrast, FTS_a is committed to its cap, so it has sufficient permits and is not charged by the government. The FTSs' profits are equal to:

$$\pi_{uOFSC_{VC}} = \left(p_{uOFSC_{VC}} - c_u\right) q_{uOFSC_{VC}} - \left|\psi y_{uOFSC_{VC}}\right| = \left(p_{uOFSC_{VC}} - c_u\right) q_{uOFSC_{VC}} + \psi y_{uOFSC_{VC}} \tag{14}$$

$$\pi_{aOFSC_{VC}} = \left(p_{aOFSC_{VC}} - c_a\right) q_{aOFSC_{VC}} \tag{15}$$

The equilibrium values of the FTSs' prices, demands, and profits with respect to the government's decision variables (γ and ψ) are presented in Eqs. (A-11) and (A-13) in the online supplementary material, if scenario SC_{VC} is realized. The equilibrium demands of the carbon offset regulation for other scenarios are also provided in Eqs. (A-1)–(A-6) in the online supplementary material. As mentioned in "Problem description," scenario SC_{VC} induces constraints $y_{uOFSC_{VC}} \le 0$ and $y_{aOFSC_{VC}} \ge 0$.

By replacing the equilibrium outputs in relations (A-11) and (A-13) into relation (11), the government's social welfare function with respect to its decision variables is as follows:

$$\begin{aligned} Maximize\ SW(\gamma,\psi) = {} & \lambda_1\left(\pi^*_{uOFSC_{VC}}(\gamma,\psi) + \pi^*_{aOFSC_{VC}}(\psi)\right) \\ & + \lambda_2\left(\int_{p^*_{uOFSC_{VC}}(\psi)}^{\frac{A_u+\beta_p\ p^*_{aOFSC_{VC}}(\psi)}{\alpha_{pu}}} \left(A_u - \alpha_{pu}\, x + \beta_p\, p^*_{aOFSC_{VC}}(\psi)\right) dx \right. \\ & \left. + \int_{p^*_{aOFSC_{VC}}(\psi)}^{\frac{A_a+\beta_p\ p^*_{uOFSC_{VC}}(\psi)}{\alpha_{pa}}} \left(A_a - \alpha_{pa}\, y + \beta_p\, p^*_{uOFSC_{VC}}(\psi)\right) dy \right) \\ & - \lambda_3 \varnothing \left(e_{u0} \times q^*_{uOFSC_{VC}}(\psi) + e_{a0} \times q^*_{aOFSC_{VC}}(\psi)\right)^2 \end{aligned} \tag{16}$$

subject to:

$$\text{Equations (A} - 11\text{) and (A} - 13\text{)} \tag{17}$$

$$\text{Equations (A} - 1\text{)} - \text{(A} - 6\text{)} \tag{18}$$

$$y_{uOFSC_{VC}}(\gamma,\psi) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{VC}}(\psi) \le 0$$

and

$$y_{aOFSC_{VC}}(\gamma,\psi) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{VC}}(\psi) \ge 0 \tag{19}$$

$$y_{uOFSC_{VV}}(\gamma,\psi) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{VV}}(\psi) \ge 0$$

or

$$y_{aOFSC_{VV}}(\gamma,\psi) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{VV}}(\psi) \ge 0 \tag{20}$$

$$y_{uOFSC_{CV}}(\gamma,\psi) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{CV}}(\psi) \le 0$$

or

$$y_{aOFSC_{CV}}(\gamma,\psi) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{CV}}(\psi) \ge 0 \tag{21}$$

$$y_{uOFSC_{CC}}(\gamma) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{CC}} \le 0$$

or

$$y_{aOFSC_{CC}}(\gamma) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{CC}} \le 0 \tag{22}$$

$$q^*_{uOFSC_{VC}}(\psi) \ge 0 \tag{23}$$

$$q^*_{aOFSC_{VC}}(\psi) \ge 0 \tag{24}$$

Equation (16) is provided to maximize the government's social welfare function according to the arguments presented in "Formulation of the costs, carbon caps, demands, and social welfare function." Constraint (17) represents the FTSs' best-response reactions toward the government decisions in scenario SC_{VC}. Constraint (18) represents the FTSs' best-response reactions toward the government decisions in other scenarios, i.e., SC_{VV}, SC_{CV}, and SC_{CC}. Constraint (19) guarantees that scenario SC_{VC} is realized. Constraint (20) ensures that scenario SC_{VV} is not realized in the carbon offset regulation. Based on the definition of SC_{VV}, both FTS_u and FTS_a violate their updated caps. So, constraint $y_{uOFSC_{VV}}(\gamma,\psi) \ge 0$ or constraint $y_{aOFSC_{V,V}}(\gamma,\psi) \ge 0$ ensures that at least one FTS has surplus permits, and scenario SC_{VV} is not realized. Similarly, constraints (21) and (22) guarantee that scenarios SC_{CV} and SC_{CC} are not realized, respectively. Constraints (23) and (24) guarantee that the equilibrium values of the FTSs' demands are positive for the realized equilibrium (SC_{VC}).

Binding carbon cap-and-trade regulation

According to the backward induction methodology, the equilibrium price, demand, and profit of each FTS are obtained based on the Nash competition mentioned in stage 3. Then, according to stage 2, the value of s_{trade} is calculated considering the Nash bargaining process under the binding cap-and-trade regulation. Finally, the government's optimization problem to maximize the social welfare function, subject to the equilibrium outputs of stages 2 and 3, is solved.

Based on the cap-and-trade regulation, the FTSs can trade permits with each other through the bargaining process. Accordingly, they can update their initial caps allocated by the government. In this regard, FTS_a sells its surplus permits, and FTS_u purchases those permits to compensate for its shortage of permits. The surplus permits under scenario k for the FTSs are as follows:

$$y_{uTRk} = \bar{s}_u(\gamma) + s_{trade} - e_{u0} \times q_{uTRk} \tag{25}$$

$$y_{aTRk} = \bar{s}_a(\gamma) - s_{trade} - e_{a0} \times q_{aTRk} \tag{26}$$

According to "Problem description," the government designs its binding cap-and-trade regulation so that scenario SC_{CC} is the realized equilibrium. Based on the definition of SC_{CC}, both FTSs are committed to their updated caps. Hence, the FTSs are not charged by the government, and only the permit fee $F \times s_{trade}$ is transferred from FTS_a to FTS_u. So the FTSs' profits under the binding cap-and-trade regulation are equal to:

$$\pi_{uTRSC_{CC}} = \left(p_{uTRSC_{CC}} - c_u\right) q_{uTRSC_{CC}} - F \times s_{trade} \tag{27}$$

$$\pi_{aTRSC_{CC}} = \left(p_{aTRSC_{CC}} - c_a\right) q_{aTRSC_{CC}} + F \times s_{trade} \tag{28}$$

The equilibrium values of the FTSs' prices, demands, and profits with respect to the government's decision variables are presented in Eqs. (A-14) and (A-16) in the online supplementary material, if scenario $SC_{C,C}$ is realized. The equilibrium demands of the cap-and-trade regulation for other scenarios are provided in Eqs. (A-7)–(A-10) and (A-30) in the online supplementary material. According to the definition of the binding cap-and-trade regulation mentioned in "Problem description," scenario $SC_{C,C}$ induces constraints $y_{uTRSC_{CC}} \geq 0$ and $y_{aTRSC_{CC}} \geq 0$. Under the binding cap-and-trade regulation, the equilibrium value of the carbon traded between the FTSs with respect to the government's decision variables (γ and ψ) is presented in Eq. (A-27) in the online supplementary material, if scenario $SC_{C,C}$ is realized.

The government's decisions should allow scenario SC_{CC} to be realized if a trade occurs (binding cap-and-trade) and scenario SC_{VC} to be realized if no trade occurs (carbon offset). In addition, in order for scenario SC_{CC} to be the final realized equilibrium, the regulation is designed so that the FTSs' profits under binding cap-and-trade (scenario SC_{CC}) are higher than those under carbon offset (scenario SC_{VC}). The social welfare function with respect to the government's decision variables is obtained in relation (29), after rewriting relation (11) according to relations (A-14), (A-16), and (A-27). The government optimizes its social welfare function based on the optimization problem presented in relations (29)–(45).

$$\begin{aligned} Maximize\ SW(\gamma,\psi) = {} & \lambda_1 \left(\pi^*_{uTRSC_{CC}}(\gamma) + \pi^*_{aTRSC_{CC}}(\gamma)\right) \\ & + \lambda_2 \left(\int_{p^*_{uTRSC_{CC}}}^{\frac{A_u + \beta_p\, p^*_{aTRSC_{CC}}}{\alpha_{pu}}} \left(A_u - \alpha_{pu}\, x + \beta_p\, p^*_{aTRSC_{CC}}\right) dx \right. \\ & \left. + \int_{p^*_{aTRSC_{CC}}}^{\frac{A_a + \beta_p\, p^*_{uTRSC_{CC}}}{\alpha_{pa}}} \left(A_a - \alpha_{pa}\, y + \beta_p\, p^*_{uTRSC_{CC}}\right) dy \right) \\ & - \lambda_3 \varnothing \left(\gamma \times e_0 \times \left(Q_{u0} + Q_{a0}\right)\right)^2 \end{aligned} \tag{29}$$

subject to:

Equations (A − 14), (A − 16), and (A − 27) (30)

Equations (A − 1) − (A − 10) (31)

Equations (A − 13) and (A − 30) (32)

$$y_{uOFSC_{VC}}(\gamma,\psi) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{VC}}(\psi) \leq 0$$

and

$$y_{aOFSC_{VC}}(\gamma,\psi) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{VC}}(\psi) \geq 0 \tag{33}$$

$$y_{uOFSC_{VV}}(\gamma,\psi) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{VV}}(\psi) \geq 0$$

or

$$y_{aOFSC_{VV}}(\gamma,\psi) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{VV}}(\psi) \geq 0 \tag{34}$$

$$y_{uOFSC_{CV}}(\gamma,\psi) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{CV}}(\psi) \leq 0$$

or

$$y_{aOFSC_{CV}}(\gamma,\psi) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{CV}}(\psi) \geq 0 \tag{35}$$

$$y_{uOFSC_{CC}}(\gamma) = \bar{s}_u(\gamma) - e_{u0} \times q^*_{uOFSC_{CC}} \leq 0$$

or

$$y_{aOFSC_{CC}}(\gamma) = \bar{s}_a(\gamma) - e_{a0} \times q^*_{aOFSC_{CC}} \leq 0 \tag{36}$$

$$y_{uTRSC_{CC}}(\gamma,\psi) = \bar{s}_u(\gamma) + s_{trade}(\gamma,\psi) - e_{u0} \times q^*_{uTRSC_{CC}} \geq 0$$

and

$$y_{aTRSC_{CC}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{CC}}\geq 0 \quad (37)$$

$$y_{uTRSC_{VV}}(\gamma,\psi)=\bar{s}_u(\gamma)+s_{trade}(\gamma,\psi)-e_{u0}\times q^*_{uTRSC_{VV}}(\psi)\geq 0$$

or

$$y_{aTRSC_{VV}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{VV}}(\psi)\geq 0 \quad (38)$$

$$y_{uTRSC_{VC}}(\gamma,\psi)=\bar{s}_u(\gamma)+s_{trade}(\gamma,\psi)-e_{u0}\times q^*_{uTRSC_{VC}}(\psi)\geq 0$$

or

$$y_{aTRSC_{VC}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{VC}}(\psi)\leq 0 \quad ((39)$$

$$y_{uTRSC_{CV}}(\gamma,\psi)=\bar{s}_u(\gamma)+s_{trade}(\gamma,\psi)-e_{u0}\times q^*_{uTRSC_{CV}}(\psi)\leq 0$$

or

$$y_{aTRSC_{CV}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{CV}}(\psi)\geq 0 \quad (40)$$

$$\pi^*_{uTRSC_{CC}}(\gamma)+\pi^*_{aTRSC_{CC}}(\gamma)>\pi^*_{uOFSC_{VC}}(\gamma,\psi)+\pi^*_{aOFSC_{VC}}(\psi) \quad (41)$$

$$q^*_{uOFSC_{VC}}(\psi)\geq 0 \quad (42)$$

$$q^*_{aOFSC_{VC}}(\psi)\geq 0 \quad (43)$$

$$q^*_{uTRSC_{CC}}\geq 0 \quad (44)$$

$$q^*_{aTRSC_{CC}}\geq 0 \quad (45)$$

Equation (29) is provided to maximize the government's social welfare function according to the arguments presented in "Formulation of the costs, carbon caps, demands, and social welfare function." Constraint (30) represents the FTSs' best-response reactions toward the government's decisions in the realized scenario, i.e., SC_{CC}. Constraints (31) and (32) represent the FTSs' equilibrium demands in other scenarios for the carbon offset and binding cap-and-trade, respectively. Similar to the arguments in "Carbon offset regulation, constraints (33)–(36) guarantee that scenario SC_{VC} is realized if carbon offset occurs. Constraints (37)–(40) guarantee that scenario SC_{CC} is realized if binding cap-and-trade occurs. Constraint (41) ensures that the FTSs' profits under the binding cap-and-trade are higher than ones in the carbon offset, i.e., SC_{CC} is the finally realized equilibrium scenario. The proof of Constraint (41) is available in the online supplementary material. Constraints (42) and (43) guarantee that the equilibrium values of the FTSs' demands are positive for SC_{VC} scenario in the carbon offset. Finally, relations (44) and (45) guarantee that the FTSs' demands for the realized equilibrium SC_{CC} are positive. It is worth to note that the two relations are not the constraints of the optimization problem since $q^*_{uTRSC_{CC}}$ and $q^*_{aTRSC_{CC}}$ are not functions of the government's decisions. Hence, the two relations are the feasibility conditions, and are based on the parameters of the problem.

Non-binding carbon cap-and-trade regulation

According to the backward induction methodology, the equilibrium price, demand, and profit of each FTS are obtained based on the Nash competition mentioned in stage 3. Then, according to stage 2, the value of s_{trade} is calculated considering the Nash bargaining process under the non-binding cap-and-trade regulation. Finally, the government's optimization problem to maximize the social welfare function, subject to the equilibrium outputs of stages 2 and 3, is solved.

In the non-binding cap-and-trade, the FTSs trade permits with each other through bargaining similar to the binding cap-and-trade. The surplus permits of the FTSs for the non-binding cap-and-trade can be calculated based on relations (25) and (26). According to "Problem description," the government designs its non-binding cap-and-trade regulation so that scenario SC_{VV} is the realized equilibrium. Based on the definition of SC_{VV}, both FTSs violate their updated caps. Hence, FTS_u and FTS_a are charged by the government, and they have to pay penalties of $\left|\psi y_{uTRSC_{VV}}\right|$ and $\left|\psi y_{aTRSC_{VV}}\right|$, respectively, for their extra emissions. So the FTSs' profits under the non-binding cap-and-trade regulation are equal to:

$$\pi_{uTRSC_{VV}}=\left(p_{uTRSC_{VV}}-c_u\right)q_{uTRSC_{VV}}-F\times s_{trade}+\psi\, y_{uTRSC_{VV}} \quad (46)$$

$$\pi_{aTRSC_{VV}}=\left(p_{aTRSC_{VV}}-c_a\right)q_{aTRSC_{VV}}+F\times s_{trade}+\psi\, y_{aTRSC_{VV}} \quad (47)$$

The equilibrium values of the FTSs' prices, demands, and profits with respect to the government's decision variables (γ and ψ) are presented in Eqs. (A-28) and (A-30) in the supplementary material, if scenario SC_{VV} is realized. According to the definition of the non-binding cap-and-trade regulation mentioned in "Problem description," scenario SC_{VV} induces constraints $y_{uTRSC_{VV}}\leq 0$ and $y_{aTRSC_{VV}}\leq 0$. Under the non-binding cap-and-trade regulation, the equilibrium value of the carbon traded between the FTSs with respect to the government's decision variables (γ and ψ) is presented in Eq. (A-37) in the supplementary material, if scenario SC_{VV} is realized.

The government's decisions should allow scenario SC_{VV} to be realized if non-binding cap-and-trade occurs and scenario SC_{VC} to be realized if carbon offset occurs. In addition, in order for scenario SC_{VV} to be the final realized equilibrium,

the regulation is designed so that the FTSs' profits under non-binding cap-and-trade (scenario SC_{VV}) are higher than those under carbon offset (scenario SC_{VC}). The social welfare function with respect to the government's decision variables is obtained in relation (48), after rewriting relation (11) according to relations (A-28), (A-30), and (A-37). The government optimizes its social welfare function based on the optimization problem presented in relations (48)–(64).

$$\begin{aligned} Maximize\ SW(\gamma,\psi) = {} & \lambda_1\left(\pi^*_{uTRSC_{VV}}(\gamma,\psi)+\pi^*_{aTRSC_{VV}}\left(\gamma,\psi\right)\right) \\ & +\lambda_2\left(\int_{p^*_{uTRSC_{VV}}(\psi)}^{\frac{A_u+\beta_p\ p^*_{aTRSC_{VV}}(\psi)}{\alpha_{pu}}}\left(A_u-\alpha_{pu}\,x+\beta_p\,p^*_{aTRSC_{VV}}(\psi)\right)dx\right. \\ & \left.+\int_{p^*_{aTRSC_{VV}}(\psi)}^{\frac{A_a+\beta_p\ p^*_{uTRSC_{VV}}(\psi)}{\alpha_{pa}}}\left(A_a-\alpha_{pa}\,y+\beta_p\,p^*_{uTRSC_{VV}}(\psi)\right)dy\right) \\ & -\lambda_3\varnothing\left(e_{u0}\times q^*_{uTRSC_{VV}}(\psi)+e_{a0}\times q^*_{aTRSC_{VV}}(\psi)\right)^2 \end{aligned} \tag{48}$$

subject to:

Equations (A − 28), (A − 30), and (A − 37) (49)

Equations (A − 1)–(A − 10) (50)

Equations (A − 13) and (A − 16) (51)

$$y_{uOFSC_{VC}}(\gamma,\psi)=\bar{s}_u(\gamma)-e_{u0}\times q^*_{uOFSC_{VC}}(\psi)\le 0$$

and

$$y_{aOFSC_{VC}}(\gamma,\psi)=\bar{s}_a(\gamma)-e_{a0}\times q^*_{aOFSC_{VC}}(\psi)\ge 0 \tag{52}$$

$$y_{uOFSC_{VV}}(\gamma,\psi)=\bar{s}_u(\gamma)-e_{u0}\times q^*_{uOFSC_{VV}}(\psi)\ge 0$$

or

$$y_{aOFSC_{VV}}(\gamma,\psi)=\bar{s}_a(\gamma)-e_{a0}\times q^*_{aOFSC_{VV}}(\psi)\ge 0 \tag{53}$$

$$y_{uOFSC_{CV}}(\gamma,\psi)=\bar{s}_u(\gamma)-e_{u0}\times q^*_{uOFSC_{CV}}(\psi)\le 0$$

or

$$y_{aOFSC_{CV}}(\gamma,\psi)=\bar{s}_a(\gamma)-e_{a0}\times q^*_{aOFSC_{CV}}(\psi)\ge 0 \tag{54}$$

$$y_{uOFSC_{CC}}(\gamma)=\bar{s}_u(\gamma)-e_{u0}\times q^*_{uOFSC_{CC}}\le 0$$

or

$$y_{aOFSC_{CC}}(\gamma)=\bar{s}_a(\gamma)-e_{a0}\times q^*_{aOFSC_{CC}}\le 0 \tag{55}$$

$$y_{uTRSC_{VV}}(\gamma,\psi)=\bar{s}_u(\gamma)+s_{trade}(\gamma,\psi)-e_{u0}\times q^*_{uTRSC_{VV}}(\psi)\le 0$$

and

$$y_{aTRSC_{VV}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{VV}}(\psi)\le 0 \tag{56}$$

$$y_{uTRSC_{VC}}(\gamma,\psi)=\bar{s}_u(\gamma)+s_{trade}(\gamma,\psi)-e_{u0}\times q^*_{uTRSC_{VC}}(\psi)\ge 0$$

or

$$y_{aTRSC_{VC}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{VC}}(\psi)\le 0 \tag{57}$$

$$y_{uTRSC_{CV}}(\gamma,\psi)=\bar{s}_u(\gamma)+s_{trade}(\gamma,\psi)-e_{u0}\times q^*_{uTRSC_{CV}}(\psi)\le 0$$

or

$$y_{aTRSC_{CV}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{CV}}(\psi)\ge 0 \tag{58}$$

$$y_{uTRSC_{CC}}(\gamma,\psi)=\bar{s}_u(\gamma)+s_{trade}(\gamma,\psi)-e_{u0}\times q^*_{uTRSC_{CC}}\le 0$$

or

$$y_{aTRSC_{CC}}(\gamma,\psi)=\bar{s}_a(\gamma)-s_{trade}(\gamma,\psi)-e_{a0}\times q^*_{aTRSC_{CC}}\le 0 \tag{59}$$

$$\pi^*_{uTRSC_{VV}}(\gamma,\psi)+\pi^*_{aTRSC_{VV}}(\gamma,\psi)>\pi^*_{uOFSC_{VC}}(\gamma,\psi)+\pi^*_{aOFSC_{VC}}(\psi) \tag{60}$$

$$q^*_{uOFSC_{VC}}(\psi)\ge 0 \tag{61}$$

$$q^*_{aOFSC_{VC}}(\psi)\ge 0 \tag{62}$$

$$q^*_{uTRSC_{VV}}(\psi)\ge 0 \tag{63}$$

$$q^*_{aTRSC_{VV}}(\psi)\ge 0 \tag{64}$$

Equation (48) is provided to maximize the government's social welfare function according to the arguments presented in "Formulation of the costs, carbon caps, demands, and social welfare function." Constraint (49) represents the FTSs' best-response reactions toward the government decisions in the realized scenario, i.e., SC_{VV}. Constraints (50) and (51) represent the FTSs' equilibrium demands in other scenarios for the carbon offset and non-binding cap-and-trade, respectively. Similar to the arguments in "Carbon offset regulation," constraints (52)–(55) guarantee that scenario SC_{VC} is realized if no trade occurs (carbon offset). Constraints (56)–(59) guarantee that scenario SC_{VV} is realized if a trade occurs (non-binding cap-and-trade). Constraint (60) ensures that the FTSs' profits under the non-binding cap-and-trade are higher than ones in the carbon offset, i.e., SC_{VV} is the finally realized equilibrium scenario. The proof of Constraint (60) is available in the online supplementary material. Constraints (61)–(64) guarantee that the equilibrium values of

the FTSs' demands are positive for the realized equilibrium in the carbon offset and non-binding cap-and-trade.

An experimental analysis

An experimental analysis is provided to illustrate the applicability of the defined problem and the proposed theoretical models. The most related work to the present article is the research conducted by Tamannaei et al. (2021b). Similar to the present article, Tamannaei et al. (2021b) investigated the competition between road and intermodal road-rail systems in the context of government intervention. Nevertheless, the present article is novel because different government regulations to curb carbon emissions are addressed. The present article designs optimal carbon offset and carbon cap-and-trade regulations, while Tamannaei et al. (2021b) investigated a government tax policy. By processing data from a case study provided by Tamannaei et al. (2021b), the parameters of the present study are estimated, and the two proposed carbon regulations are compared with the tax policy. The details of estimation are provided in the online supplementary material. Table 2 presents the estimated parameters of the models.

The present article also compares the energy efficiency indicator of the carbon regulations from different government perspectives. The Energy Efficiency Indicator (EEI) is equal to total cargoes hauled by the FTSs per their total fuel consumptions (Patterson 1996). Relation (65) calculates the indicator.

$$EEI = \frac{Total\ output}{Total\ energy\ consumption} = \frac{q_{ujk} + q_{ajk}}{\theta_1 D_u q_{ujk} + \theta_1 (1-d) D_a q_{ajk} + \theta_2 d D_a q_{ajk}} \quad (65)$$

The equilibrium results of the carbon regulations

As mentioned in "Theoretical background," a carbon regulation is effective when it provides less total carbon emissions than status quo. Moreover, a regulation is defined to be optimal to implement if it is effective and provides higher social welfare than the other effective regulations. An energy-efficient regulation improves the value of the energy efficiency indicator (EEI) compared to the status quo.

We calculate the equilibrium results based on the government perspectives under the status quo, carbon tax, carbon offset, binding cap-and-trade, and non-binding cap-and-trade regulations. The government perspectives considered in the experimental analysis are economic, social, and environmental ones. If the government considers a specific perspective, it supposes the related importance coefficient equal to 0.9, and the other two coefficients equal to 0.05. Accordingly, the three perspectives are considered in the social welfare function and the dominant perspective impressively affects that function. For example, for the government's environmental perspective, we have $\lambda_1 = 0.05$, $\lambda_2 = 0.05$, and $\lambda_3 = 0.9$. In all mentioned carbon regulations, the FTSs determine their equilibrium prices and demands based on the Bertrand-Nash competition.

Economic perspective

In the online supplementary material, Table (A-1) presents the equilibrium outputs under different carbon regulations from the economic perspective of the government. Given a specific criterion, suppose that regulation b is preferable to regulation a. Relation $a \tilde{<} b$ represents this preference. Also, $a \equiv b$ denotes the indifference between regulations a and b, given the specific criterion. According to the values of total carbon emissions (TCE^*) in Table (A-1), all carbon

Table 2 The main parameters and their values in the experimental analysis

Parameter	Explanation	Value
A_u (A_a)	Market base of FTS_u (FTS_a)	1700 tons/day (1650 tons/day)
c_u (c_a)	Fuel cost imposed on FTS_u (FTS_a)	4.753 \$/ton (9.450 \$/ton)
e_{u0} (e_{a0})	Average value of CO_2 emissions produced by FTS_u (FTS_a) for carrying one unit of demand from origin to destination	0.123568 $\frac{\text{ton } CO_2}{\text{ton}}$ (0.050690 $\frac{\text{ton } CO_2}{\text{ton}}$)
$c_{u,\,tax}$ ($c_{a,\,tax}$)	Fuel cost and carbon tax imposed on FTS_u (FTS_a) in the carbon tax regulation	34.219 \$/ton (23.141 \$/ton)
Q_{u0} (Q_{a0})	The demand of freight transportation for FTS_u (FTS_a) in the status quo	1055.8 ton (1031.7 ton)
F	Carbon trading price	800 $\frac{\$}{\text{ton } CO_2}$
$\varnothing$	Environmental damage of one unit of CO_2 emission	50 $\frac{\$}{\text{ton } CO_2}$
α_{pu} (α_{pa}) and β_p	Self-price and cross-price sensitivities of demands	$\alpha_{p,\,u} = \alpha_{p,\,a} = 1.190$ $\beta_p = 0.476$
α	Relative bargaining power of FTS_u	0.45

regulations are capable to reduce total carbon emissions produced by the FTSs in comparison to the status quo. The preference of carbon regulations in terms of TCE^* and Social Welfare (SW^*) criteria in the economic perspective are as follows:

TCE^* criterion: Status quo $\tilde{<}$ Carbon tax $\tilde{<}$ Binding cap-and-trade $<$ Non-binding cap-and-trade $<$ Carbon offset

SW^* criterion: Carbon tax $\tilde{<}$ Status quo $\tilde{<}$ Binding cap-and-trade $<$ Non-binding cap-and-trade $<$ Carbon offset.

According to the definition of the energy efficiency indicator, the values of the indicator are calculated for the carbon regulations under the economic perspective of the government, as shown in Fig. 5. Although all carbon regulations improve the energy efficiency indicator, the carbon offset regulation is superior to the other carbon regulations.

Social perspective

Table (A-2) presents the equilibrium outputs under different carbon regulations from the social perspective of the government. According to Table (A-2), carbon offset and non-binding cap-and-trade are not effective regulations to reduce total carbon emissions, as they cannot reduce total carbon emissions compared to the status quo. So, the carbon tax regulation is compared with the binding cap-and-trade. Since the values of TCE^* and SW^* in the binding cap-and-trade are lower than those in the carbon tax, a comparison between the two mentioned carbon regulations is quite sophisticated. For convenient comparison, we calculate the values of carbon taxes (or subsidies) imposed on (or paid to) the road and rail systems so that total carbon emissions produced by the FTSs are equal to those of binding cap-and-trade. Accordingly, the comparison between the carbon tax and binding cap-and-trade regulations is feasible with respect to SW^*.

There are two procedures so that the value of TCE^* produced by the FTSs is equal under carbon tax and binding cap-and-trade regulations. First, based on the Bertrand-Nash competition, if the fuel tax imposed on the road system equals 2.142 \$/*lit*, and the subsidy paid to the rail system equals 0.1 \$/*lit*, the FTSs produce the same carbon emissions as those of the binding cap-and-trade. Second, if the fuel tax imposed on both the road and rail systems equals 2.08 \$/*lit*, the FTSs also produce the same carbon emissions as those of the binding cap-and-trade. Accordingly, the total value of the carbon tax imposed on the two mentioned systems is equal to 0.8 \$/kg CO_2. Table A-3 in the online supplementary material indicates the equilibrium results of the carbon tax-and-subsidy, carbon tax, and binding cap-and-trade regulations considering the equal TCE^* for the regulations under the government's social perspective.

According to Table A-3, the binding cap-and-trade regulation provides higher SW^* than the carbon tax regulation regarding the equal TCE^*. The order of carbon regulations in terms of TCE^* and SW^* in the social perspective is as follows:

TCE^* criterion: Status quo ≡ Non-binding cap-and-trade ≡ Carbon offset $\tilde{<}$ Carbon tax≡ Carbon tax-and-subsidy ≡ Binding cap-and-trade

SW^* criterion: Carbon tax $\tilde{<}$ Carbon tax-and-subsidy $\tilde{<}$ Binding cap-and-trade $\tilde{<}$ Status quo ≡ Non-binding cap-and-trade ≡ Carbon offset

According to the definition of the energy efficiency indicator, the values of the indicator are calculated for the carbon regulations under the social perspective of the government, as shown in Fig. 6. Carbon offset and non-binding cap-and-trade regulations cannot improve the value of the energy efficiency indicator. In contrast, carbon tax-and-subsidy, carbon tax, and binding cap-and-trade regulations enhance the

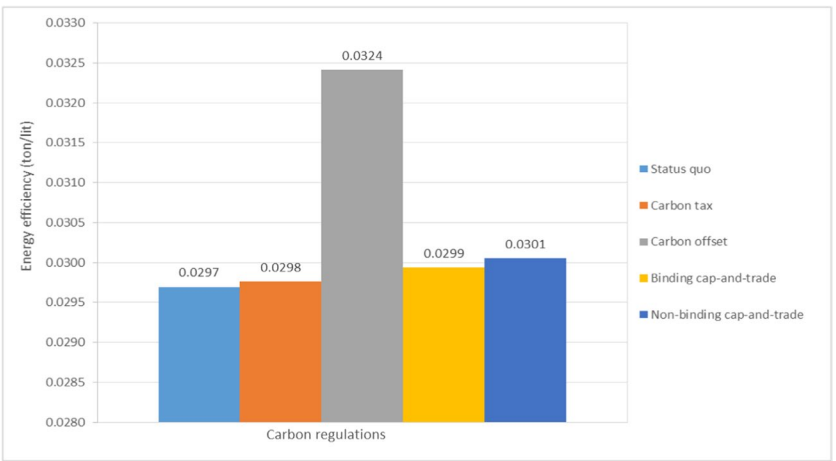


Fig. 5 The values of the energy efficiency indicator for the carbon regulations from the economic perspective

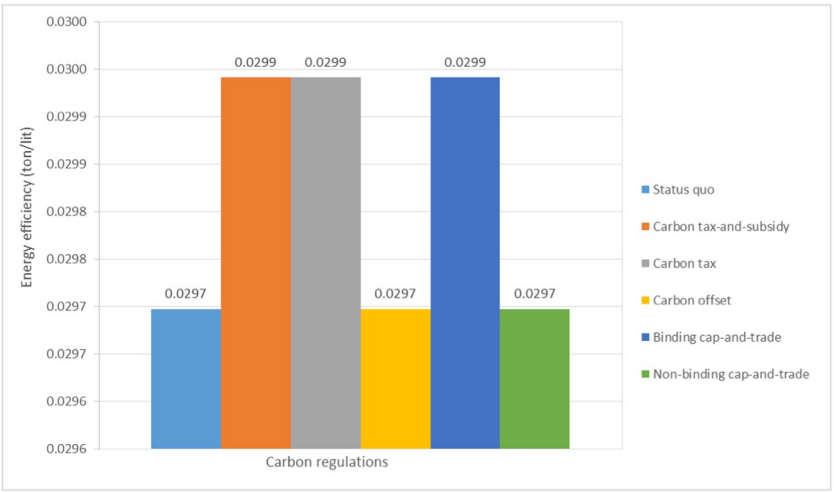


Fig. 6 The values of energy efficiency indicator for the carbon regulations from the social perspective

energy efficiency indicator so that the value of the indicator is equal for the three mentioned carbon regulations.

Environmental perspective

Table (A-4) presents the equilibrium outputs under different carbon regulations from the environmental perspective of the government. According to the values of TCE^* in Table (A-4), all carbon regulations are capable to reduce total carbon emissions produced by the FTSs in comparison to the status quo. The order of carbon regulations in terms of TCE^* and SW^* in the environmental perspective is as follows:

TCE^* criterion: Status quo $<$ Carbon tax $<$ Binding cap-and-trade $<$ Non-binding cap-and-trade $<$ Carbon offset

SW^* criterion: Status quo $<$ Carbon tax $<$ Binding cap-and-trade $<$ Non-binding cap-and-trade $<$ Carbon offset

According to the definition of the energy efficiency indicator, the values of the indicator are calculated for the carbon regulations under the environmental perspective of the government shown in Fig. 7. Although all carbon regulations improve the energy efficiency indicator, the carbon offset regulation is superior to the other carbon regulations.

Sensitivity analysis

Under cap-and-trade and carbon offset regulations, allocating initial caps and determining the penalty of violation are significant variables affecting followers' decisions (Chen 2023; Kim and Yu 2018; Lin and Jia 2017; Narassimhan et al. 2018; Xu et al. 2016a; Zhang et al. 2019). The initial caps allocated to the enterprises should be in a sense that total carbon emissions decreases after trading permits. In other words, carbon emissions produced by the enterprises should be less than their updating caps. Some enterprises may violate their updating caps at the end of the period or present incorrect reports of their carbon emissions to increase their profits (Anand and Giraud-Carrier 2020; Pan et al. 2019). Thus, the penalty of violations made prevents from those

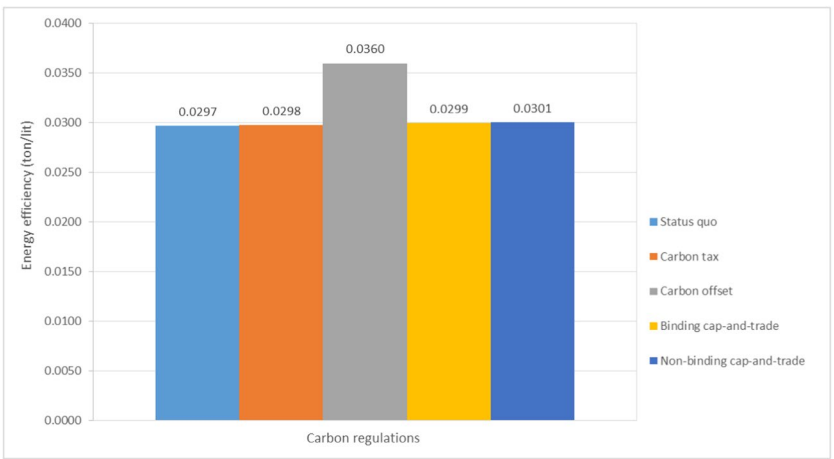


Fig. 7 The values of energy efficiency indicator for the carbon regulations from the environmental perspective

violations. Accordingly, from different government perspectives, the effects of γ and ψ are investigated on the social welfare function under cap-and-trade and carbon offset regulations.

According to Fig. 8, various values of ψ do not affect the social welfare function under the binding cap-and-trade from different government perspectives. Moreover, based on Fig. 8(a), if the value of γ reduces in the feasible region, i.e., interval [0.962,1], the value of the social welfare function increases. Accordingly, the equilibrium results are $\gamma = 0.962$ and $\psi = 947.2$. Although dotted curves or lines help to illustrate the general schematic of charts, they do not exist in the feasible region. Moreover, solid black lines indicate the border(s) of the feasible region created by solid curves or lines. The green or black solid points refer to the feasible region which are composed of only one point. In addition, the green point indicates the equilibrium result.

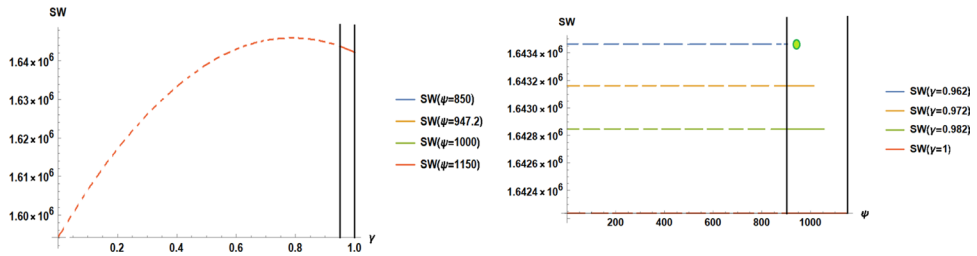
According to Fig. 9, if the non-binding cap-and-trade is the effective regulation, i.e., from economic and environmental perspectives, when the values of γ and ψ simultaneously increase in the feasible region, the social welfare function increases. For example, if the value of γ is equal to 0.67, when ψ increases, the social welfare decreases. However, if the value of γ is equal to 0.87, when ψ increases, the social welfare increases. Accordingly, the equilibrium results are $\gamma = 0.946$ and $\psi = 1143.4$.

According to Fig. 10, under the carbon offset from the economic perspective, if the values of γ and ψ simultaneously increase in the feasible region, the social welfare function increases. Based on Fig. 10(a), whether the value of ψ increases or also γ increases, the growth rate of the social welfare increases. Based on Fig. 10(b), whatever the value of γ increases, the reduction range of the curve decreases. Accordingly, the equilibrium result equals to $\gamma = 1$ and $\psi = 4367.1$.

According to Fig. 11, under the carbon offset from the environmental perspective, if the values of γ and ψ simultaneously increase in the feasible region, the social welfare function increases. Based on Fig. 11(a), an increase of ψ has a more positive effect on improving social welfare than an increase of γ. Based on Fig. 11(b), the value of ψ is significant to find the equilibrium result. Accordingly, the equilibrium result equals to $\gamma = 0.649$ and $\psi = 8098.1$.

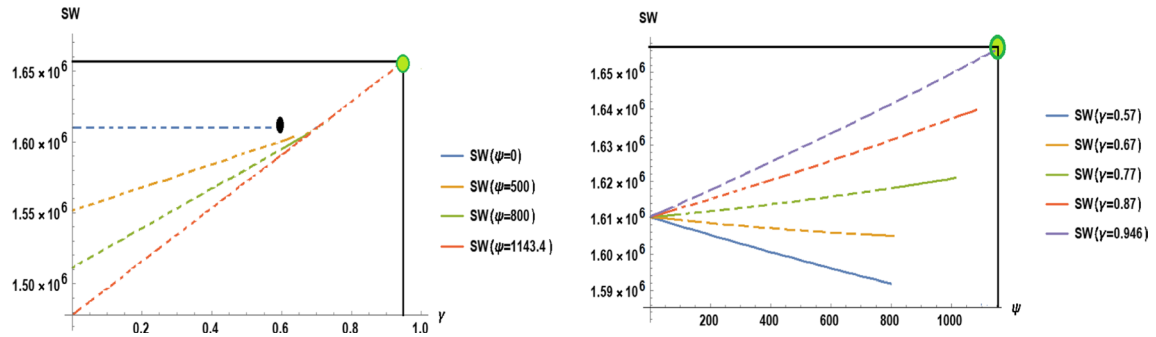
Discussions and implications

In each of the following paragraphs, based on the experimental analysis, we discuss the effectiveness, optimality, and energy efficiency of a regulation among carbon offset, binding cap-and-trade, non-binding cap-and-trade, and carbon tax. As defined in "Problem description," the binding nature of a cap-and-trade regulation refers to the FTSs'



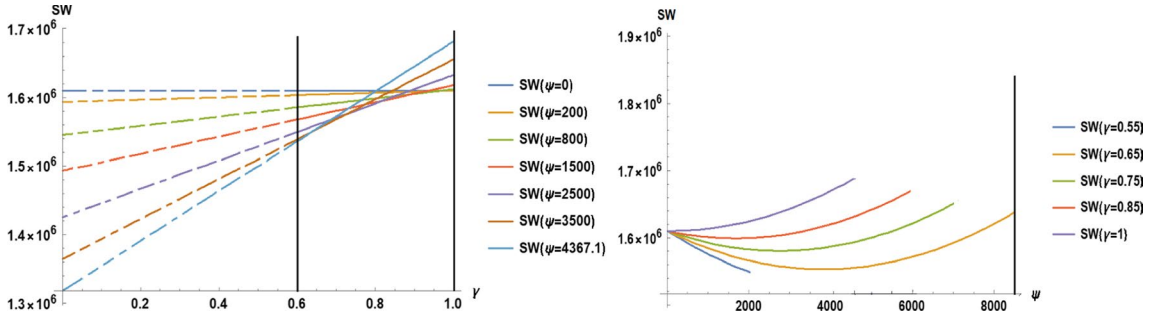
(a) The effect of γ (b) The effect of ψ

Fig. 8 The effects of γ and ψ on *SW* under the binding cap-and-trade from economic, social, and environmental perspectives



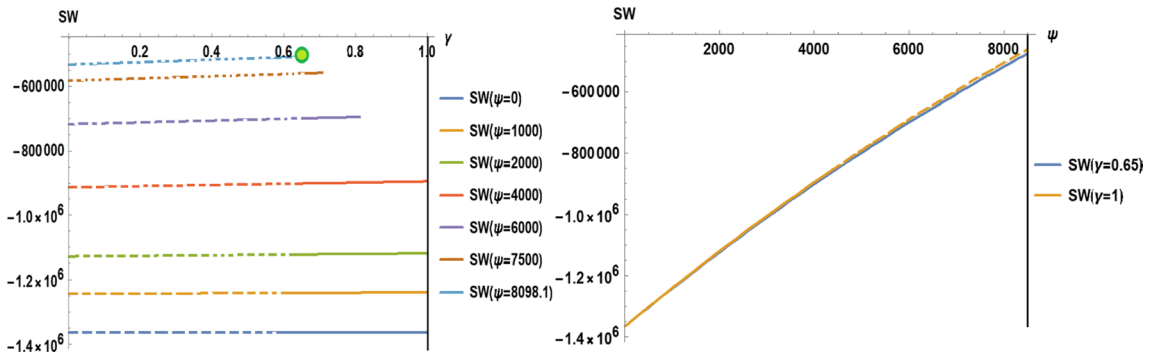
(a) The effect of γ (b) The effect of ψ

Fig. 9 The effects of γ and ψ on *SW* under the non-binding cap-and-trade from economic and environmental perspectives



(a) The effect of γ (b) The effect of ψ

Fig. 10 The effects of γ and ψ on *SW* under the carbon offset from the economic perspective



(a) The effect of γ (b) The effect of ψ

Fig. 11 The effects of γ and ψ on *SW* under the carbon offset from the environmental perspective

Table 3 The effects of the government perspective on the effectiveness, optimality, and energy efficiency of the carbon regulations

Dominant government perspective	The effectiveness status of regulations					Optimal regulation	Energy-efficient regulation(s)
	Carbon tax-and-subsidy	Carbon tax	Carbon offset	Binding cap-and-trade	Non-binding cap-and-trade		
Economic	-	✓	✓	✓	✓	Carbon offset	Carbon offset
Social	✓	✓	×	✓	×	Binding cap-and-trade	Binding cap-and-trade, tax-and-subsidy, and tax
Environmental	-	✓	✓	✓	✓	Carbon offset	Carbon offset

commitment to their updated caps. Table 3 summarizes this comprehensive discussion.

Carbon offset: If the social perspective is dominant for the government, the carbon offset is not an effective regulation, in a sense that its equilibrium outputs are the same as those of the status quo. However, the environmental perspective is dominant for the government; the carbon offset is an effective regulation and is superior to the cap-and-trade and carbon tax regulations in reducing total carbon emissions. This superiority originates from the government's decision to impose a high value for the penalty of violation in the carbon offset regulation. In other words, the high value of penalty induces the FTSs' commitment to the regulation; the commitment is realized through increasing the FTSs' prices, and consequently reducing their demands. Although consumer surplus and the profit of FTS_u drop significantly under the carbon offset regulation, emission reduction (environmental improvement) is

so higher in the carbon offset than the other regulations, which results in the superiority of the carbon offset to the other ones. Consequently, carbon offset is the optimal regulation to implement from the environmental perspective of the government. Moreover, the carbon offset regulation is also optimal when the government considers the economic perspective.

From the economic and environmental perspectives, the carbon offset regulation provides the highest value of the energy efficiency indicator among other carbon regulations. Thus, the carbon offset is the optimal and energy-efficient regulation to implement from both perspectives.

Binding and non-binding cap-and-trade: If the social perspective of the government is dominant, the carbon offset and non-binding cap-and-trade are ineffective regulations. In this regard, the optimal regulation is selected from the carbon tax and binding cap-and-trade. The findings show that the binding cap-and-trade provides higher social welfare than the carbon tax and carbon tax-and-subsidy when the mentioned regulations lead to an equal value of total carbon emissions. Therefore, the binding cap-and-trade is the optimal regulation to implement from the social perspective of the government. Moreover, the binding cap-and-trade provides the highest value of the energy efficiency indicator from the social perspective. Thus, the regulation is optimal and energy-efficient from the social perspective.

From the economic and environmental perspectives the non-binding cap-and-trade is not only an effective regulation, but also it makes the FTSs committed to their updated caps. Accordingly, the non-binding cap-and-trade converges into a new binding cap-and-trade. More specifically, the amounts of FTSs' violations of their updated caps are equal to zero in the non-binding cap-and-trade. The findings show that the binding cap-and-trade allocates higher caps to the FTSs than the mentioned non-binding cap-and-trade, while the latter imposes a higher value for the penalty of violation than the former. In other words, the two mentioned regulations make the FTSs committed to their updated caps through different approaches. On the other hand, the non-binding cap-and-trade improves the energy efficiency compared to the status quo.

Carbon tax: The carbon tax is an effective regulation from the economic, social, and environmental perspectives of the government, respectively. The carbon tax regulation also increases the energy efficiency indicator for the mentioned perspectives, while it is not the optimal regulation to implement from any of those perspectives.

Some studies have generally concluded that the cap-and-trade regulation is optimal to implement (Chen et al. 2020; Chen and Nie 2020; Esmaeili et al. 2020; Ji et al. 2019; Tang and Yang 2020; Xu et al. 2016b; Zhang et al. 2017). In line with the literature, the experimental analysis indicates that the cap-and-trade regulation is optimal from the social perspective of the government. However, these results contribute to the knowledge in the literature by revealing that the carbon offset regulation is optimal under the economic and environmental perspectives of the government.

The obtained results are aligned with the literature addressing the penalty of violation and can extend it. We can suppose two comparisons. First, under the economic and environmental perspectives, our experimental analysis indicates that the penalty of violation in the non-binding cap-and-trade (ψ^*=1143.4) is higher than that in the binding cap-and-trade (ψ^*=947.2). Second, the carbon offset regulation designed from the environmental perspective imposes higher penalty of violation (ψ^*=8098.1) than the one from the economic perspective (ψ^*=4367.1). In line with the literature, a regulation with a higher penalty of violation induces a higher value for the proportion of demands of FTS_a to demands of FTS_u (i.e., $\frac{q_a^*}{q_u^*}$), where FTS_a adopts the green technology. In comparison 1, the mentioned proportion under the non-binding cap-and-trade is $\frac{1016.1}{982.4} = 1.034$, while the binding cap-and-trade induces the proportion of $\frac{1020.8}{1004.4} = 1.016$. In comparison 2, the mentioned proportion under the carbon offset in the environmental perspective is $\frac{1155.7}{482.5} = 2.395$, while the regulation in the economic perspective induces the proportion of $\frac{1098.6}{748} = 1.469$. According to the experimental analysis, it is concluded that if a penalty of violation increases, the value of the mentioned proportion increases. This is aligned with Zhang et al. (2019) that concluded the penalty of violation has a positive correlation with green technology adoption.

However, the equilibrium results contribute to the literature by revealing that increasing the penalty of violation may affect the FTSs' profits in a *different* manner. In comparison 1, transition from binding cap-and-trade to non-binding cap-and-trade increases the penalty of violation. The transition simultaneously increases both the FTSs' profits (π_a^*: 953,702 > 945,171 , π_u^*: 922,616 > 918,923). However, comparison 2 provides a different conclusion. Transition in the carbon offset regulation from economic perspective to the environmental perspective increases the penalty of violation. The transition increases the profit of FTS_a ($\pi_a^* : 1,122,388 > 1,014,130$), but decreases the profit of FTS_u ($\pi_u^* : 683,291 < 873,906$). The results indicate that the equilibrium profit of FTS_a is positively correlated with the penalty of violation. However, the relationship between the equilibrium profit of FTS_u and the penalty of violation is not straightforward and depends on the carbon regulation. If the cap-and-trade (carbon offset) regulation is implemented, the relationship is positive (negative).

In response to Questions 1 and 2 posed in "Theoretical gaps and research questions," we present a new definition for the effectiveness and optimality of carbon regulations.

A carbon regulation is effective if it provides less total carbon emissions than the status quo. Moreover, a regulation is optimal to implement if it is effective and generates higher social welfare than the other effective carbon regulations. In response to Question 3, the multimodal road-rail system produces less carbon emissions than those of its rival, i.e., road system, and plays a significant role to reduce total carbon emissions in the transportation industry. Moreover, the mentioned system can increase the value of the net government utility from its social and environmental considerations.

The results from the experimental analysis can respond to Questions 4 and 5. From the economic and environmental perspectives of the government, carbon tax, carbon offset, and carbon cap-and-trade regulations are effective; among them, the carbon offset is the optimal regulation. From the social perspective of the government, carbon tax and binding cap-and-trade regulations are effective; among them, the binding cap-and-trade is the optimal regulation. Hence, from the social perspective of the government, the development of a marketplace infrastructure for trading carbon permits is justifiable because the corresponding regulation, i.e., the cap-and-trade, is optimal.

The results can be applied to assess the two hypotheses mentioned in "Theoretical background." First, according to Figs. 5, 6 and 7, if a carbon regulation is effective, it improves the energy efficiency indicator more than the status quo, so H1 is supported. In other words, the optimal carbon regulation is also the energy-efficient regulation from different government perspectives. Second, based on Table 3, whether H2 is supported or rejected depends strongly on the government perspective. On the one hand, cap-and-trade is not an optimal regulation to implement from the economic and environmental perspectives. Hence, H2 is rejected when these two perspectives are dominant for the government. On the other hand, H2 is supported when the social perspective of the government is dominant.

Some practical implications are recommended based on the experimental analysis. These implications can help policymakers to adopt an optimal carbon regulation concerning the government's perspective. On the one hand, we suggest a government with economic or environmental perspectives to implement the carbon offset regulation. This is because the carbon offset regulation provides the highest (lowest) value for the total profit of FTSs (environmental damage) from the economic (environmental) perspective among other effective carbon regulations. On the other hand, we suggest a government with social perspective to implement the binding cap-and-trade regulation. This is because the binding cap-and-trade regulation provides the highest value for the total profit of FTSs from the social perspective among other effective carbon regulations, while the total carbon emissions of these carbon regulations are the same.

Conclusions

The present study contributes to the literature on carbon regulations by addressing a competitive market consisting of road and multimodal road-rail freight transportation systems (FTSs), and designing carbon cap-and-trade, carbon offset, and carbon tax regulations for this market. For the first time, the present article takes into account the heterogeneity of transportation systems to reduce carbon emissions regarding economic, social, and environmental aspects. In this regard, this article concentrates on the competition between road and multimodal road-rail freight transportation systems; the latter can be considered as an environmentally sustainable system.

Methodologically, the study moves beyond the existing literature on carbon regulations by developing a sequential game-theoretic model for the competitive market. In the upper level, the government, as the Stackelberg leader, sets the carbon emission reduction coefficient and the penalty of violation for a carbon regulation to maximize a social welfare function containing economic, social, and environmental dimensions. In the lower level, a Nash game is developed to analyze duopoly pricing competition in the freight transportation market. In the carbon cap-and-trade regulation, there is also a medium level in which the FTSs trade carbon permits through a bargaining process. Under the carbon offset and cap-and-trade regulations, we propose a new procedure for allocating the initial carbon caps to the FTSs, which is a hybrid of both grandfathering and benchmark methods. The advantage of the proposed procedure is to benefit the multimodal road-rail as the sustainable green system, in a sense that it always allocates a lower (higher) carbon cap to the road (multimodal road-rail) system compared to the status quo.

We suggest two cap-and-trade regulations based on the FTSs commitment or violation to their updated caps, which are binding and non-binding cap-and-trade. The experimental analysis reveals that under certain conditions, the non-binding cap-and-trade reduces total carbon emissions and converges into a new binding cap-and-trade. The conditions refer to the domination of economic and environmental perspectives of the government. It is also worth to note that the two regulations make the FTSs committed to their updated caps through different approaches. The binding cap-and-trade induces the commitment by allocating higher caps, while the non-binding cap-and-trade induces the commitment by imposing a higher value of penalty.

The present article extends our understanding of the linkage between the optimal carbon regulation and the government perspective. On the one hand, our findings recommend that the government should implement the cap-and-trade regulation when the social perspective is dominant. Since the realization of the cap-and-trade requires a marketplace infrastructure, the development of this infrastructure is

justifiable from the social perspective of the government. On the other hand, carbon offset regulation is recommended when the dominant perspective of the government is economic or environmental. Accordingly, the development of the mentioned marketplace infrastructure is not justifiable under these two government perspectives.

Our work is not free of limitations, and some ideas are worthy to extend the research. First, this study considers road and multimodal road-rail freight transportation systems in a duopoly market. However, future research can focus on an oligopoly market with more than two agents. Second, while the study considers deterministic demands for the two transportation systems, other variables affect demand, such as the time and safety of hauled materials until they are delivered to customers. Hence, the demands of these systems can be considered stochastic in future research. Third, the carbon cap-and-trade regulation can be enriched using the auction method to allocate initial carbon caps to the freight transportation systems. This is because some scholars believe that the auction is a cost-effective method to allocate the initial carbon cap (Álvarez and André 2014; Alvarez et al. 2017).

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Data availability All the data generated or analyzed during this study are included in this article.

Declarations

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